County Vulnerability to SNAP Losses Under Expanded ABAWD Work Requirements

Evidence from Michigan's Staggered Reinstatement

Jiaming Zhang

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Department of Agricultural, Food, and Resource Economics
Michigan State University

Motivation

Policy context

- SNAP ABAWD time limits: 3 months of benefits per 36 months unless working $\geq$80 hrs/month
- States received broad waivers after 2008; waivers expired as labor markets recovered
- 2026 One Big Beautiful Bill Act (OBBBA): work requirements extend to ages **55–64**

Research question

Which Michigan counties face the greatest risk of SNAP participation losses under expanded work requirements, and what county characteristics predict larger impacts?

Michigan's Staggered Reinstatement

Four treatment cohorts:

1. **Jan 2017** — 4 counties (Kent, Oakland, Ottawa, Washtenaw)
2. **Jan 2018** — 10 counties (suburban/mid-metro)
3. **Jul 2018** — 68 counties (rural, gradual phase-in)
4. **Oct 2018** — 1 county (Wayne/Detroit)

$\Rightarrow$ **All 83 counties** eventually treated

$\Rightarrow$ Identification: *not-yet-treated* as controls

Cohort	Unemp.	Bach.
Jan 17	3.9%	40.6%
Jan 18	4.4%	27.5%
Jul 18	6.7%	18.2%
Oct 18	6.8%	22.1%

Key: Treatment timing $\sim$ labor market strength $\Rightarrow$ confounding risk

Data

- **Michigan FAP** county-month panel: Jan 2014 – Dec 2019 (83 counties $\times$ 72 months)
- **Outcome**: log(SNAP recipients per 1,000 adults 18–49)
  - Min $\approx$ 52/1k (no zeros $\Rightarrow$ log well-defined)
  - Mean = 225/1k; SD = 94
- **County characteristics**: BLS unemployment, ACS poverty & education, CDC SVI, USDA RUCC
- **Estimation sample**: 5,229 county-month obs after trimming anticipation window

Empirical Strategy

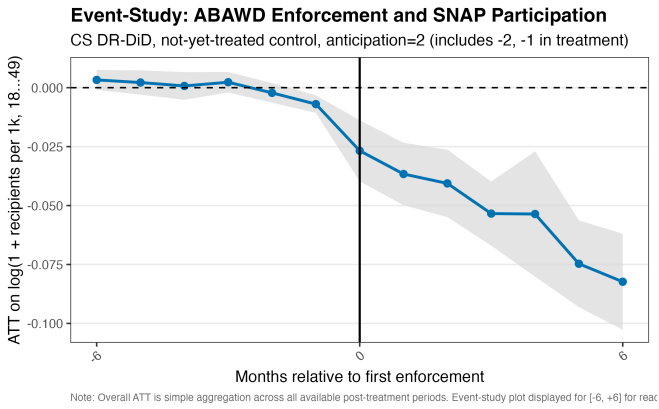
Callaway & Sant'Anna (2021) staggered DiD:

$$ATT(g,t) = \mathbb{E}\big[Y_t(g) - Y_t(0) \mid G = g\big]$$

- Control group: *not-yet-treated* counties
- Doubly-robust (DR) with unemployment rate as covariate
- Anticipation $\delta = 2$ months (waiver removal announced $\sim$4 months early)
- Event study: $e \in [-6, +6]$ months
- Clustering: county level

Placebo: DDD comparing adult recipients (affected) vs. child recipients (unaffected)

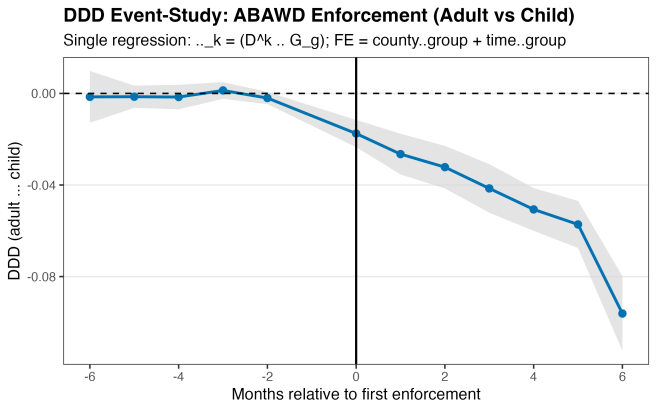
Main Result: Event Study



Key findings:

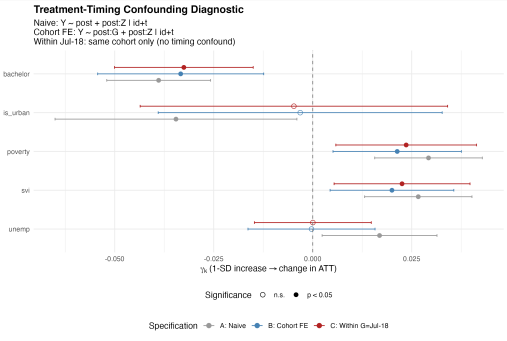
- Pre-trends flat ($e \leq -3$)
- Overall ATT $= -0.060$ (SE $= 0.010$, $p < 0.01$)
- $\approx$ **5.8% reduction** in SNAP participation
- Effects accumulate over 6 months (recertification phase-in)

DDD Placebo: Adult vs. Child Recipients



- Adult recipients (subject to ABAWD): significant decline
- Child recipients (not subject): **no effect**
- Rules out county-level confounders affecting both groups

Heterogeneity: Treatment-Timing Confounding



Diagnostic: A. Naive → B. Cohort FE → C. Within Jul-18

Spurious (vanish in B & C):

- Unemployment: $+0.016^{**} \rightarrow 0.000$
- Urbanicity: $-0.034^{**} \rightarrow -0.003$

Genuine (survive all specs):

- Bachelor's share: -0.033^{***}
- Poverty rate: $+0.021^{**}$
- SVI: $+0.020^{**}$

LASSO: selects only bachelor's & poverty

Projection: 2026 OBBBA Expansion to Ages 55–64

County-specific predicted ATT $\widehat{ATT}_c = \overline{ATT} + \hat{\delta}_1 \tilde{Z}_c^{\text{bach}} + \hat{\delta}_2 \tilde{Z}_c^{\text{pov}}$

Projected loss $\widehat{\Delta Y}_c = \widehat{ATT}_c \times S_c^{55\text{-}64} \times \bar{Y}_{c,\text{pre}}$ Uncertainty: bootstrap $B = 2{,}000$

Three scenarios:

- Baseline (current unemployment)
- Recession (+3 pp unemployment)
- Tight labor (−2 pp unemployment)

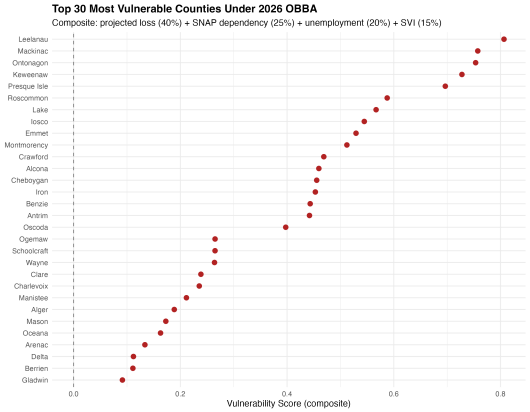
Three channels:

- Compliance: harder to meet 80-hr req
- Enrollment: more people on SNAP
- Waiver: >10% unemp ⇒ exemption

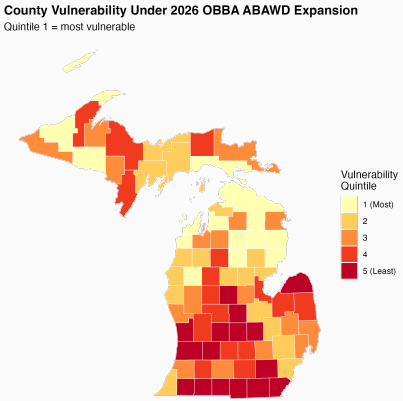
Projection Results

Scenario	Total loss
Baseline	$-9{,}918$
	[7,200 – 12,600]
Recession ($+3$ pp)	$-13{,}440$
Tight labor (-2 pp)	$-7{,}805$

- $\sim$**10,000** individuals projected to lose benefits
- Recession amplifies by 36%
- Recovery attenuates by 21%



County Vulnerability Map



Most vulnerable:

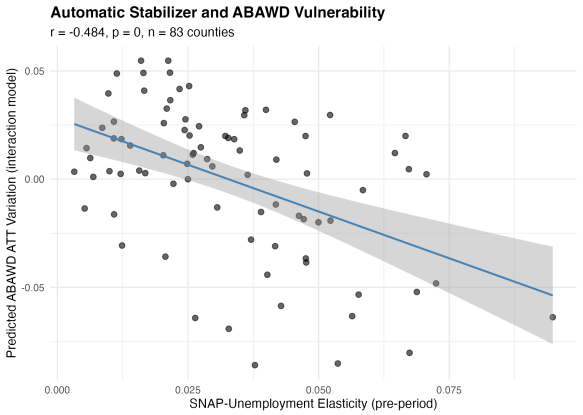
- Northern Lower Michigan
- Upper Peninsula
- High baseline SNAP + moderate education

Least vulnerable:

- SE Michigan metro
- Strong labor markets + service infrastructure

Rank uncertainty: median width = 12 positions (bootstrap)

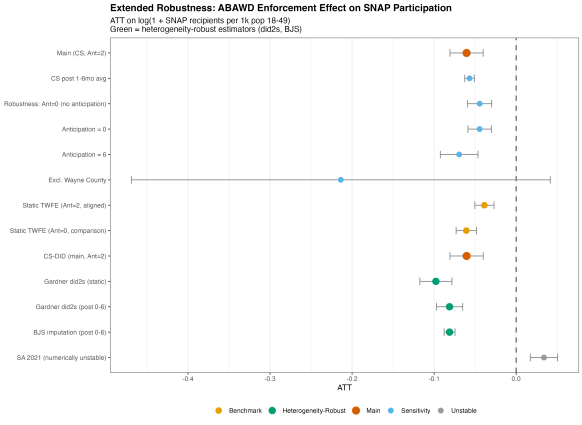
SNAP as Automatic Stabilizer



- Pre-period: SNAP enrollment rises with unemployment ($\bar{\beta} = 0.033$)
- Work requirements weaken this stabilizer
- $r = -0.48$ ($p < 0.001$): counties where SNAP was most countercyclical $\Rightarrow$ largest participation losses

Implication: ABAWD requirements bind hardest where the safety net is most needed

Robustness



Estimator agreement:

- CS-DID: -0.060
- did2s: -0.080
- BJS imputation: -0.080
- Stacking: -0.051
- SA: $+0.034$ (rank deficiency)

Inference:

- HonestDiD $\bar{M}^* > 2.0$
- Fisher permutation $p < 0.002$
- Two-way clustering: SE unchanged

Policy Implications

1. **Targeted outreach**: Vulnerability ranking $\Rightarrow$ allocate E&T slots, case management, and community partnerships by projected need
2. **Administrative capacity**: Gradual phase-in $\Rightarrow$ counties need advance staffing for recertification surge
3. **Age-specific design**: 55–64 population faces greater compliance barriers (lower LFP, health limitations, retraining barriers) $\Rightarrow$ consider age-adjusted thresholds, expanded medical exemptions
4. **Countercyclical concern**: Work requirements weaken SNAP's automatic stabilizer function precisely where it is most needed

Summary

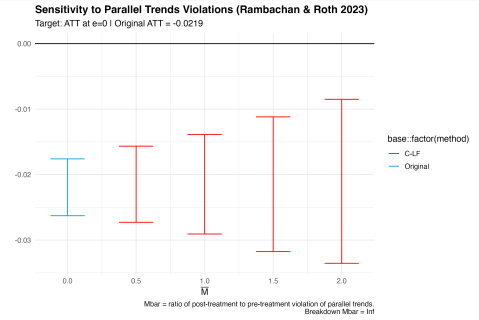
1. ABAWD enforcement $\Rightarrow$ **5.8% reduction** in SNAP participation (gradual, 6-month phase-in)
2. Heterogeneity by unemployment & urbanicity is spurious (treatment-timing confounding); **genuine** heterogeneity through education & poverty
3. 2026 OBBBA expansion: $\sim$**10,000** projected losses statewide (CI: 7,200–12,600); concentrated in northern MI & Upper Peninsula
4. County vulnerability ranking provides **actionable tool** for implementation planning

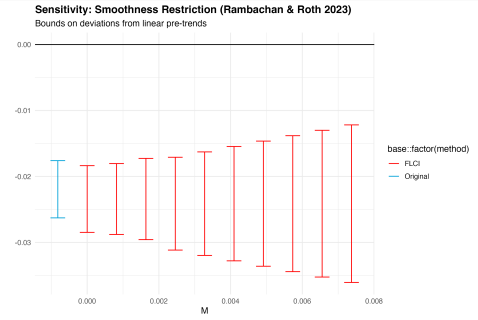
Thank you! Questions?

zhangji6@msu.edu

Backup Slides

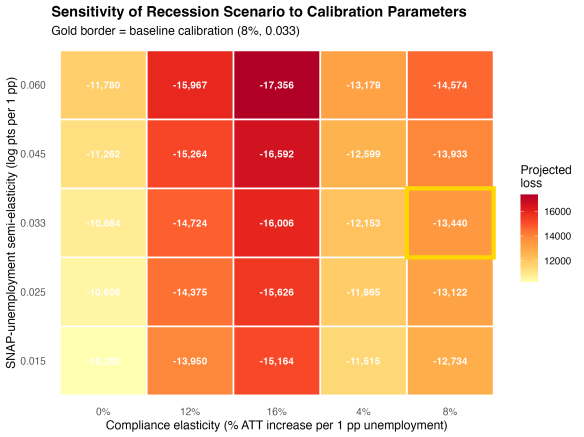
HonestDiD Sensitivity





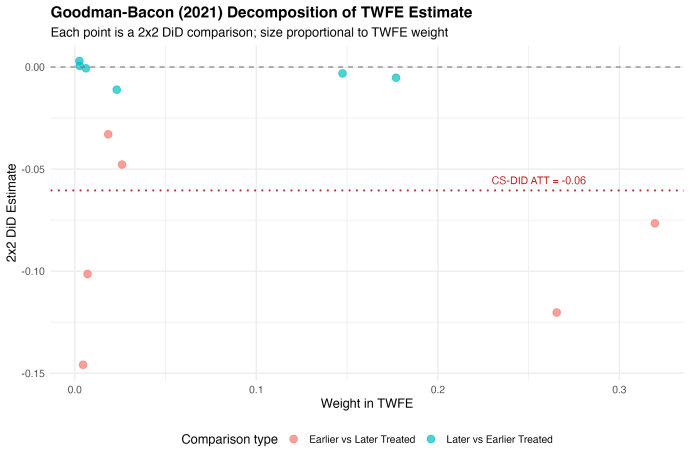
Breakdown value $\bar{M}^* > 2.0$: conclusion holds even if post-treatment violations are $2\times$ the largest pre-treatment violation.

Sensitivity of Recession Scenario



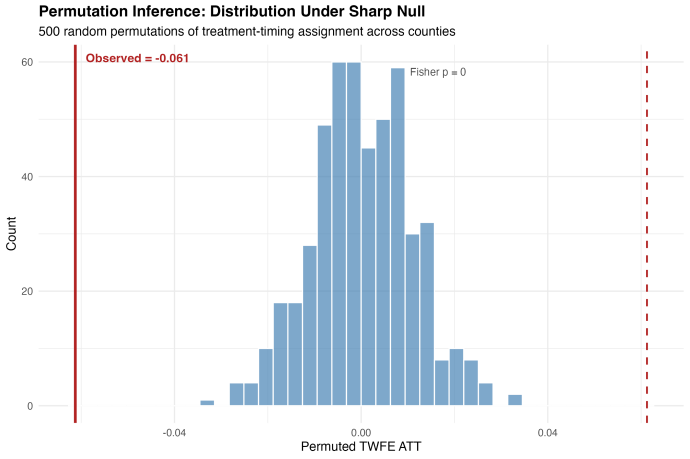
Gold cell = baseline (8% compliance elast., 0.033 semi-elast.). Grid: 10,300–17,400.

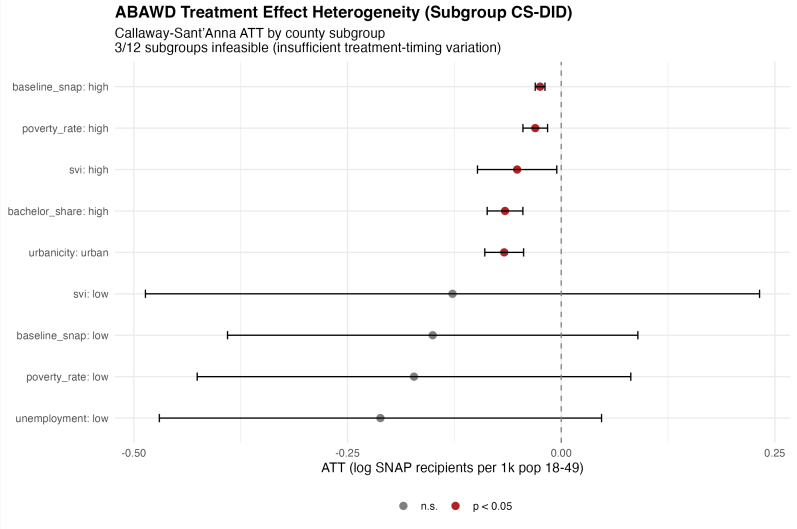
Goodman-Bacon Decomposition



Earlier-vs-later (wt=0.64): -0.093. Later-vs-earlier (wt=0.36): -0.005. No sign flip $\Rightarrow$ limited TWFE bias concern.

Permutation Inference



500 Fisher randomization draws. Observed ATT (-0.061) far in left tail; $p < 0.002$.

Heterogeneity Forest Plot